AD7190

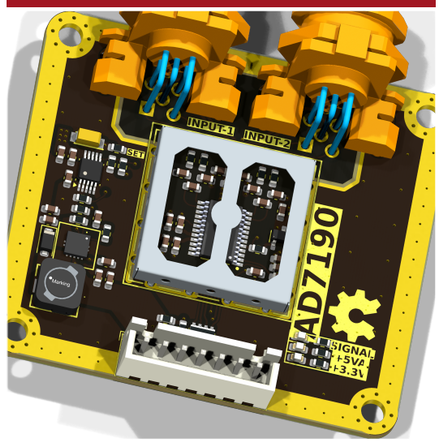
Variant:

2026-10-24

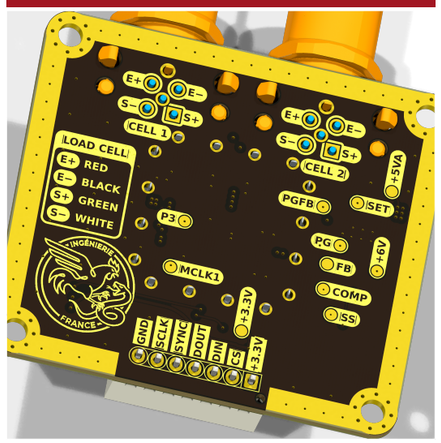
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

OPEN_WEIGHT - AD7190 24-bit load-cell front-end, part of OPEN THRUST LAB.
Single +3.3 V host supply, SPI interface, 2 x 4-wire bridge inputs.

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 24-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
+3.3V host -> TPS61086
6 V -> LT3042 +5VA
(AVDD + excitation)

DEBUG NOTE:
TP on every rail.
Wait >= 600 ms (PG) before ADC init.

DESIGN NOTE:
SPI mode 3,
SCLK <= 5 MHz,
RDY on DOUT.

DESIGN NOTE: REFIN1 = bridge excitation: ratiometric.

LAYOUT NOTE: AFE under the WE-SHC shield H5, boost SW node away.

Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: ea7f156
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: OPEN_WEIGHT - AD7190	File Name: OPEN_WEIGHT.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 9